FYP Handbook

FACULTY OF ENGINEERING
INFORMATION TECHNOLOGY UNIVERSITY
LAHORE, PAKISTAN

Contents

Introduction

- Each FYP group will have maximum of THREE (3) student members. In special scenarios, there can be a third from cross-department (after approval of FYP coordinator and Department Chairperson) bringing an extra skill to enhance the scope of the project.

- ITU Faculty of Engineering will share list of FYPs proposed by faculty members and Industry collaborators before the semester start.

- Students can either register any of proposed FYPs or they can register their own FYP idea after getting it approved from advisor and co-advisor.

- Either Advisor or Co-advisor must be from Faculty of Engineering. However, it is mandatory to have your FYP Advisor from Faculty of Engineering (EE/CE).

- Electrical Engineering Projects must have hardware/ practical implementation module. Software/Algorithm only projects will not be accepted in general.

FYP forms:

- Every group must submit the hard copy of FYP form signed by the advisor, Co-advisor and FYP Coordinator for Chairperson Signature before the deadline (1st week of 7th semester).
- Every group must upload the scan copy of signed FYP form on FYP Google Classroom before the deadline (1st week of 7th semester).

FYP Approvals:

- Every group should make a short presentation of maximum three minutes to justify the project titles in accordance with FYP Forms (2nd week of 7th semester).
- FYP Coordinators, Department Chairperson and other members from Faculty of Engineering will evaluate presentation and approve the Projects.

- **FYP proposal defense**
  - FYP proposal defense will be held in 6th week of 7th semester. Department will announce the date, time and venue of proposal defense.
  - Every group will upload a signed FYP proposal report by advisor and co-advisor on google classroom before proposal defense. Proposal report should contain introduction, motivation, literature review, block diagrams showing flow & components of FYPs, FYP timelines, and task assignment to each group member.
  - I grade will be assigned to groups failed to defend FYP proposals and submit proposal reports.

- **FYP mid evaluation**
  - FYP mid evaluation will be conducted in 1st week of 8th semester. Mid evaluation will be marked on the basis of demo of prototype and presentation. Department will announce the date, time and venue of prototype demo and presentations.

- Every group will upload a signed mid FYP report by advisor and co-advisor on google classroom before mid-evaluation. Report should contain details of components, experimental setup, results, prototype etc.
- I grade will be assigned to groups failed to pass mid-evaluation and submit mid FYP reports.

- **FYP final defense**
    - FYP final defense will be held in 15th week of 8th semester and will be evaluated by FYP evaluation committee consists of advisor, co-advisor and a third evaluator.
    - Department will nominate the 3rd evaluator for each FYP group. Department will announce the date, time and venue of proposal defense and project display.
    - Every group will upload a signed FYP report by advisor and co-advisor on google classroom before final presentation.
    - Grades will be assigned to groups according to their performance. I and F grades can be assigned. In case of F grade, students will register FYP-I again.
    - In case of I grade, student will spend next few months on project completion.

- **Weekly log book**: Every group will upload FYP log on google classroom on weekly basis and share it with your advisor through email.
  In your weekly log, you should record the *FYP progress* you have made during week or thought about in comprehensive manner. This includes for example ideas, new concept/terms learnt, designs, sketches, summaries of reports/journals that you have read (including full details of the source of the report), records of meetings with supervisors or other individuals from whom you are seeking advice etc.
  Enter everything into the book in a disciplined and reasonably neat fashion. This means that you will be encouraged to think and write in a disciplined and thus efficient manner. However, a real log book is not a 'work of art' and will always have crossings out, rough sketches rather than finished drawings and so on.
  You should record meetings with your supervisor in your log book including agenda, meeting minutes and follow-up work. This may also include the queries, responses and further planning.
  There is good reason for recording information fully in this way. In general, you are never able to work full-time on any one job and there will always be other distractions. It is not surprising that the retention time of information is not long, and it is frustrating to have to go over ground that has already been covered. Spending that little time in writing up will thus be a good investment. A good record will enable you to use your time efficiently as well as to provide the best base for writing up your project later, which is usually done under considerable pressure to meet a deadline.

    - Every group is required to update weekly logbook Google document on weekly basis
    - Each group should create a single google document for FYP weekly Logbook. Share this google document with your advisor, co-advisor and FYP group members
    - Enter the link of this google document in google sheet (link will be provided by FYP.EE)
    - FYP weekly logbook Word template will be provided, you need to create Google document using the provided template. You can copy the template in

the same google document to create log of every week. Logs in Google document should be in ascending order i.e., Week 1 log, Week 2 log, Week 3 log etc

- Week numbering should be semester week numbering, for example if you will write your first log in week 3 of semester, week # 3 should be entered in logbook not week 1
- Name the google word document as Group#_SupervisorName_FYP Weekly Logbook.docx
- In every week's logbook, every group member must summarize his weekly progress, problem faced with steps toward solutions, plan for next week.
- Summarize in concise but comprehensive manners, use proper engineering and scientific terms. Don't use general language i.e, we did literature review, write your conclusions from readings, concepts and terms learnt etc.
- You suppose to meet your Advisor every week. It is your duty to plan meeting with your advisor. You need to summarize the meeting with your supervisor in every week's logbook in term of agenda, minutes, future plans/tasks decide and so on.
- If your advisor is busy, ask him to assign his RA/graduate student for weekly meeting.
- You should complete every week's logbook by Sunday, TA will review your logbook after each week deadline (Sunday) and will mark your logbook on weekly basis.

- **Meetings**: Meetings are another important aspect of project work. ***For this project you are obliged to have regular meetings with your FYP Advisor on average half an hour per week.*** These meetings may be video conferences, phone calls as well as face-to-face. It is also very important to keep contact with your advisors through emails. There are other times when you are likely to be involved in meetings like with other academic or research staff, meetings with professionals outside the university environment. Benefit could come from meetings with other students with whom there is a commonality of work. Whoever it is, the people from whom you are seeking help or information will almost certainly be busy with many tasks and responsibilities, so their knowledge, help and time must be fully utilised.

  You shouldn't infer from this that you should approach busy people sparingly. Generally, experts in a subject will be highly motivated to help a young newcomer and will nearly always respond in an enthusiastic way, as long as you communicate courteously, knowledgeably and in a professional manner. However, it is usual to make an appointment or to arrange a meeting in advance. You cannot normally expect to get the best attention without giving some notice. In such meetings you are the recipient of the help, and it is then appropriate that planning the meeting and making it work be your responsibility. This means that you must plan meetings carefully and in advance to optimise the communication. You should plan an agenda which might have the following items: any update of information or progress from your previous meeting; the efficient briefing of your partner-in-meeting about the background of your requests; the information you want to get out of the meeting; and the timing of a future meeting. ***Use your logbook to record agendas, minutes and follow-up work.***

  Note that the regular meetings should be recorded in the FYP Weekly Logbooks.

- **Hard copies**:

    - Three hard copies of final FYP report must be approved and signed by advisor and co-advisor and submitted to FYP Coordinator.
    - Certificate of originality contained in thesis must be signed by students.
    - The cover page of FYP thesis hardcopy must be embossed as per provided standards. The background color of binding will be black and text and logo will be in golden color. Sample images will be provided.

- **FYP data on CD/DVD:**
    - Final FYP report in .pdf format
    - Final FYP report in .tex format
    - Proposal defense presentation
    - Mid evaluation presentation
    - Final defense presentation
    - Conference or Journal Paper
    - Poster
    - Videos showing FYP setup and working
    - FYP website link
    - FYP data folder (containing all simulation and experimental data files)

- **Reimbursements**:
    - For reimbursement of FYP expenses, a working prototype must be submitted to advisor along with all the receipts.
    - Reimbursement will not be made for faulty equipment or components.
    - A maximum of 10,000 PKR per student will be reimbursed.
    - Advisor will raise a note sheet after receiving all the receipts and ITU will reimburse the adviser who will later reimburse to students.

Timeline, Deliverables and Weightages

Event/ Activity	Time/ Deadline	Deliverables	Weightage
FYP Form Submission	7th Semester, 1st Week	FYP form signed by advisor and co-advisor • Hard copy to FYP secretary • Soft copy on classroom	Registration
FYP Approval	7th Semester, 2nd Week	• Presentation	Registration
Proposal Defense	7th Semester, 6th Week	• Presentation • Signed proposal report	• 3.5% Advisor • 1.5% Co-advisor • 3% Report
Mid Evaluation	8th Semester, 1st Week	• Presentation • Prototype demo • Signed mid FYP report	• 12% Advisor • 8% Co-advisor • 5% Report
Final Defense	8th Semester, 15th Week	• Final defense presentation • Project Demo • Signed final FYP report	• 19% Advisor • 13% Co-advisor • 13% 3rd Evaluator • 8% Report
		Three hardcopies of approved report from advisor and co-advisor FYP data on DVD containing: • Proposal, Mid, Final defense presentations • Proposal, mid, final FYP reports in .docx and pdf • Data folder including FYP data, codes, videos, poster, FYP website link (if you designed it)	2%
Weekly Logbook	7th - 8th Semester	Weekly logbook after FYP registration • Upload on google classroom weekly • Submit its copy to advisor	2%
Project Display	8th Semester, 16th Week	• Poster presentation • Project display	10%

FYP Registration Form

Information Technology University
Faculty of Engineering
FYP Registration Form

Project Title: ____

____

Advisor's Name: ____

Co- Advisor's Name: ____

Group Member Details:

Sr. No.	Registration No.	Student Name	Contact Number	Signature
1				
2				
3				

IMPORTANT NOTE: Students should ensure that the Selected FYP includes the characteristics of Complex Engineering Problems (CEPs) as per the Guidelines of HEC and PEC.

1. **Involvement of United Nation Sustainable Development Goal (SDGs) in selected FYP**

☐ No poverty
☐ Zero Hunger
☐ Good Health and well being
☐ Quality Education
☐ Gender Equality
☐ Clean water and Sanitation
☐ Affordable and Clean Energy
☐ Decent work and Economic growth
☐ Industry, Innovation and Infrastructure
☐ Reduced Inequality
☐ Sustainable cities and communities
☐ Responsible consumption & production
☐ Climate Action
☐ Life below water
☐ Life on land
☐ Peace justice and strong institutions
☐ Partnership for the goal

2. **Innovation in selected FYP**
   ☐ Novel Idea
   ☐ Modification/Upgradation of an existing idea
   ☐ Reproducing existing idea

3. **Which of the knowledge areas does your FYP incorporate. (Please refer to Page 32 of the linked HEC document.)**
   - ☐ Engineering foundation
   - ☐ Major breadth core
   - ☐ Major depth core
   - ☐ Multi-disciplinary

4. **Specify is there is any sort of Industry Academia Linkage**
   ☐ Industry Funded Project
   ☐ Industry collaborative Project
   ☐ No industry linkage

5. **Selected FYP is Research based or Funded Project**
   ☐ Part of Funded Project (write name of funding body)

   ____

   ☐ Funding planned for future (write name of funding body)

   ____

   ☐ Non funded project

Signatures:

Advisor	**Co- Advisor**
____________________	____________________
Chairperson	**FYP Coordinator**
____________________	____________________

FYP Weekly Log Book

Week Number		**Date**		**Group No.**	
Project Title					
Name of Advisor					

Work achieved this week(by individual group members)
Problems encountered
Work planned for next week(by individual group members)
Project plan status (on, ahead, behind)
Meeting with Advisor (date time of meeting, agendas, minutes, and follow-up work)

Tentative Grading Criteria

A+	90.00	100.00
A	80.00	89.99
A-	75.00	79.99
B+	70.00	74.99
B	65.00	69.99
B-	60.00	64.99
C+	55.00	59.99
C	50.00	54.99
F	**0**	**49.99**
I	**Incomplete**	

CLOs and PLOs of FYP

#	Description	C	A	P
CLO1	Describe and comprehend the fundamentals of related subjects	C2		
CLO2	i) Identify and analyze the problem statement ii) Have clear idea of the expected outcomes	C4		
CLO3	i) Know the available tools to build an intended system ii) Make a technical comparison to choose the most appropriate option			P3
CLO4	i) Report the existing literature and identify the state-of-the-art ii) Analyze the developed solution by means of simulations	C4		
CLO5	Develop a viable and reliable solution in an efficient way to meet the project objectives by following scientific and safety standards	C5		
CLO6	Relate the solution with the societal needs and demonstrate its impact and sustainability	C3	A3	
CLO7	Practice the professional ethics by: i) Not taking undue credit of someone else's work ii) Providing appropriate citations where applicable		A3	
CLO8	i) Complete the allocated task ii) Contribute strongly as a team member		A3	
CLO9	i) Compose high quality technical documents ii) Confidently demonstrate presentation skills	C2		
CLO10	i) Achieve the expected outcomes in timely fashion ii) Define the milestones, and continuously evaluate the progress	C4	A3	
CLO11	Identify relevant research and development domains in a broader context and construct on the developed solution to solve given problems	C5	A3	

FYP Program Learning Outcomes (PLOs):

	CLO1	CLO2	CLO3	CLO4	CLO5	CLO6	CLO7	CLO8	CLO9	CLO10	CLO11
PLO1 Engineering Knowledge	✓										
PLO2 Problem Analysis		✓									

PLO3 Development of Solutions					√						
PLO4 Investigation				√							
PLO5 Modern tool usage			√								
PLO6 The Engineer and Society						√					
PLO7 Environment &Sustainability						√					
PLO8 Ethics							√				
PLO9 Individual and Teamwork								√			
PLO10 Communication									√		
PLO11 Project Management										√	
PLO12 Lifelong Learning											√

FYP Proposal Report and Defense

(a) FYP Proposal Report Evaluation Rubrics

FYP Proposal Report Evaluation Rubrics							
	Components	**Marks**	**Poor**	**Satisfactory**	**Good**	**Excellent**	**CLOs Mapping**
			25%	25%-50%	50-75%	75-100%	
PR1	**Organization and Formatting**	2	The report is not in standard format and completely unorganized	The report is in standard format but structured in a confusing manner	The report is in standard format and is readable	The report is in standard format with proper headings, page numbers, captions, font size and perfectly structured	CLO9
Technical Contents							
PR2	**Problem Statement & Scope**	2	Key points are missing. The contents are hard to understand and interpret	All key points are covered but no use of charts, graphs, figures etc., to explain salient points	All key points are covered but limited use of charts, graphs, figures etc., to explain salient points	All key points are covered and effective use of charts, graphs, figures etc., to explain salient points	CLO6
PR3	**Sustainability**	2					CLO6
PR4	**Literature Review & Analysis**	3					CLO2
PR5	**Proposed Solution/ Methodology**	3					CLO5

PR6	**Milestones & Timelines**	2					CLO10
PR7	**Work Division**	2					CLO8
PR8	**Use of Language and Grammar**	2	Lot of spelling and grammatical mistakes and writing is not understandable.	Frequent spelling and grammatical mistakes and writing needs significant editing	Occasional spellings and grammatical mistakes and writing is acceptable	Almost no spelling or grammatical mistakes and excellent writing	CLO9
PR9	**Referencing/ Work Citations**	2	No or very few references and no citation in the text	Reasonable references but no citation in the text	Reasonable references and citation in the text but not in IEEE format	Reasonable references in IEEE format and well cited in the text	CLO3

(b) FYP Proposal Presentation Evaluation Rubrics

FYP Proposal Defense Presentation Evaluation Rubrics							
	Components	**Marks**	**Poor**	**Satisfactory**	**Good**	**Excellent**	**CLOs Mapping**
			0-2	2-5	5-8	8-10	
PD1	**Problem Statement & Scope**	10	Clear list of statements of problem is missing	Statements of problems are listed but does not reflect scope of project, thus need to be refined	List of statements of problem are clear and adequate for scope of project	List of statements of problem are concise and precise in relation to scope of project	CLO6
PD2	**Sustainability (UN SDG)**	10	No idea about the impact of professional engineering solutions in societal and environmental contexts under UN SDGs	Understand the impact of professional engineering solutions to some extent in societal and environmental contexts under UN SDGs	Understand the impact of professional engineering solutions in societal and environmental contexts under UN SDGs	Understand the impact of professional engineering solutions in societal and environmental contexts under UN SDGs and demonstrate knowledge of and need for sustainable development	CLO6
PD3	**Literature Review & Analysis**	10	No or little background study of problem, market, and	Organized but incomplete & inconsistent background study of problem, market,	Competent background study of problem, market, and existing solutions conducted and reported	Comprehensive and systematically executed background study of problem, market, and existing solutions	CLO2

			existing solutions	and existing solutions			
PD4	**Design Methodology/ Approach**	10	No or unclear plan for the execution of project	Basic plan is given without adequate details	Realistic plan for project with only major details covered	Realistic plan for project with minor details covered	CLO5
PD5	**Presentation and Organization Skills**	10	Contents unorganized, inability to answer *any* questions, *faltering* English used	Contents organized in order, inability to answer *some* questions accurately, *hesitant* English used	Contents organized in order and with proper details, answer *most* questions adequately, *proper* English used	Contents organized in detail and in order, able to answer *all* questions with precision, *fluent* English used	CLO9

FYP Proposal Report:

- You need to write all FYP reports (proposal, mid and final) in Latex. FYP Report Latex template will be shared with you with instructions on how to use Latex.
- You should get your proposal report review by advisor and co-advisor before Submission.
- You should email the proposal reports to advisor and co-advisor at least one week before submission deadline. This will provide enough time to advisor and co-advisor for review. While emailing proposal report to advisors, keep your group members and **fyp.eed@itu.edu.pk** / **fyp.ced@itu.ed.pk** in cc.
- Advisor and co-advisor should sign the report with date.
- You have to upload the approved and signed FYP proposal report in pdf in response to assignment that will be created on google classroom.
- Name the proposal report as Group#00_Supervisor_Name_FYP_Proposal.pdf.

<u>FYP Proposal Report Description:</u>

ABSTRACT

Abstract of the proposal

INTRODUCTION

Discuss the opening perspective of the problem area, the challenge in that area and refine the challenge into a concise

MOTIVATION

CHALLENGES

PROBLEM STATEMENT

Unmet need or Problem

What is the unmet need or problem the FYP is aiming to solve. How significant is the problem? Quantify as much as possible.

In case of a research problem, show the significance of the unsolved problem.

Who needs it

List the type of customers who want a solution to the problem. For each type of customers indicate the potential market size.

Incase of a research problem, identify where this research will be used?

SUSTAINABILITY

Summarize the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

LITERATURE REVIEW

What has been done by others to solve the problem? What solutions are already present in the market? What are their disadvantages?

In case of a research problem, literature review of the state-of-the-art should be included.

Literature Analysis

Analyze the summarized literature to come up with the proposed solution.

Give your value proposition. How is your solution going to be different and better than Literature?

In case of a research problem, how the proposed research solution is expected to be better than the state-of-the-art?

PROJECT DEVELOPMENT METHODLGY/ARCHITECTURE

Distribute the project goals into smaller objectives/modules and highlight deliverables for each objective.

Explain the modules of project through a system level block diagram.

Students may also mention tools, technologies and suitability of the method(s) to be employed with justification.

In case of a research problem, show the few approaches that will be investigated in the project?

PROJECT MILESTONES AND DELIVERABLES

Clear milestones should be defined at the start of the project which includes a Gantt chart in the project management document

WORK DIVISION

Clear work division among group members to be shown

COSTING

Cost breakdown of the project to be shown, estimating the total cost of the final product.

REFERENCES

Any material/information referred from research papers, books or/and websites should be acknowledged under this heading.

The students are recommended to use any bibliography management software to keep track of references. e.g., EndNote, Mendlay, Jabref

Tutorial Videos on Creating Gantt Charts:

https://www.youtube.com/watch?v=-crraoSMxc0

https://www.youtube.com/watch?v=ibdDmhipDOQ

FYP Proposal Defense:

- FYPs proposal defense Schedule will be shared by department.
- Every group member should join/reach the venue at least 10 minutes before scheduled time.
- Every member needs to present some portion of presentation and he/she should present in English.
- FYP proposal defense presentation will be evaluated on the basis of abovementioned Rubrics, so make sure to include at least these points in your presentation.
- You will have 10 minutes max. for your proposal presentation.
- You will be uploading your FYP Proposal Presentation after FYP Proposal Defense in response to assignment (will be created) on Google Classroom.
- You have to incorporate the changes in your presentation recommended by advisor and co-advisor before uploading it.
- Name the ppt/pptx file as Group#00_Advisor_Name_Proposal.pptx

FYP Mid Report and Defense

(a) FYP Mid Report Evaluation Rubrics

FYP Mid Report Evaluation Rubrics							
	Components	Marks	Poor	Satisfactory	Good	Excellent	CLOs Mapping
			25%	25%-50%	50-75%	75-100%	

MR1	**Organization and Formatting**	2	The report is not in standard format and completely unorganized	The report is in standard format but structured in a confusing manner	The report is in standard format and is readable	The report is in standard format with proper headings, page numbers, captions, font size and perfectly structured	CLO9
				Technical Contents			
MR2	**Updated Literature Review & Analysis**	3	Key points are missing. The contents are hard to understand and interpret	All key points are covered but no use of charts, graphs, figures etc., to explain salient points	All key points are covered but limited use of charts, graphs, figures etc., to explain salient points	All key points are covered and effective use of charts, graphs, figures etc., to explain salient points	CLO2
MR3	**Engineering Design/ Methodology**	3					CLO5
MR4	**System/Architecture Requirements**	3					CLO4
MR5	**Revised Milestones & Timelines**	2					CLO10
MR6	**Revised Work Division**	2					CLO8
MR7	**Use of Language and Grammar**	2	Lot of spelling and grammatical mistakes and writing is not understandable.	Frequent spelling and grammatical mistakes and writing needs significant editing	Occasional spellings and grammatical mistakes and writing is acceptable	Almost no spelling or grammatical mistakes and excellent writing	CLO9
MR8	**Referencing/ Work Citations**	3	No or very few references and no citation in the text	Reasonable references but no citation in the text	Reasonable references and citation in the text but not in IEEE format	Reasonable references in IEEE format and well cited in the text	CLO3

(b) FYP Mid Report Evaluation Rubrics

	Components	**Marks**	**Poor**	**Satisfactory**	**Good**	**Excellent**	**CLOs Mapping**
			0-2	2-4	5-8	8-10	
MD1	**Background, Literature Review& Analysis**	10	No or little review of background and literature	Review is conducted but incomplete and inconsistent	Sufficient review conducted but not systematic	Comprehensive and systematically executed	CLO2

MD2	**Engineering Design/ Methodology**	10	No engineering design at all	Conventional design with new approach	Novel design with minimal impact	Novel design which solves complex engineering problem	CLO5
MD3	**System/Architecture & Requirements**	5	System architecture and its requirements are too vague	System architecture and its requirements are included in an ordinary way	System architecture and its requirements are included in an acceptable way	System architecture and its requirements are included in an excellent way with details of components, setups, diagrams etc	CLO4
MD4	**Engineering Knowledge**	10	Unable to answer basic questions	Answered some of basic questions	Seems novice and can answer basic questions only	Answers to questions are strengthen by rationalization and explanation	CLO1
MD5	**Presentation and Organization Skills**	10	Contents were unorganized and delivered in unclear manner	Relevant contents were delivered in reasonable manner	Concise but not precise contents were delivered in reasonable manner	Concise and precise contents delivered in fluent manner	CLO9
MD6	**Professional Ethical Values**	10	Student is unaware of ethical implications of the work	Student is aware of ethical implications of the work but unable to ensures adherence to ethical best practices	Student is aware of ethical implications of the work and tried his best to ensures adherence to ethical best practices	Student has a good grasp on all the ethical implications of the work and ensures adherence to ethical best practices	CLO7

FYP Mid Report:

- You should modify/extend your FYP proposal report in Latex on Overleaf to prepare FYP Mid report. But you should save original .tex file of your proposal for future.
- You FYP Mid Report should have following contents (Details in attached Rubrics):
  - Updated/Revised background/literature review.
  - Analysis based on background/literature review to figure out previous approaches/methods used to solve the problem with their performance parameters. You can use comparison table, charts, graphs etc to show your comparison.
  - Engineering Design: your engineering design approach to solve the problem.
  - System architecture & requirements: Complete system design and architecture and its components and module requirements.
  - Results obtained so far, results can be simulation or experimental results. If you have fabricated some prototype, include details of it as well.
  - Updated/revised milestones and timelines.
  - Conclusion: conclude your work till date.
- You should email the mid reports to advisor and co-advisor at least one week before submission deadline. This will provide enough time to advisor and co-advisor for review. While emailing proposal report to advisors, keep your group members and fyp.eed@itu.edu.pk /fyp.ced@itu.ed.pk in cc.
- Advisor and co-advisor should sign the report with date.

- You have to upload the approved and signed FYP mid report in pdf on google classroom assignment which will be created soon.
- Deadline for submission of FYP mid report will be 1 week before FYP Mid presentations which will be held in 1st week of your 8th Semester.
- Name the proposal report as Group#00_Supervisor_Name_FYP_Mid.pdf.
- Your FYP Mid report and presentation will be evaluated on the basis of abovementioned Rubrics, so make sure to include at least these points in your report presentation.

FYP Mid Presentations:

- FYPs Mid defense Schedule will be shared by department.
- Every group member should join/reach the venue at least 10 minutes before scheduled time.
- Every member needs to present some portion of presentation and he/she should present in English.
- FYP proposal defense presentation will be evaluated on the basis of abovementioned Rubrics, so make sure to include at least these points in your presentation.
- You will have 10 minutes max. for your proposal presentation.
- You will be uploading your FYP Mid Presentation after FYP Proposal Defense in response to assignment (will be created) on Google Classroom.
- You have to incorporate the changes in your presentation recommended by advisor and co-advisor before uploading it.
- Name the ppt/pptx file as Group#00_Advisor_Name_Mid.pptx

FYP Final Defense and Report

(a) FYP Final Report Evaluation Rubrics

	Components	**Marks**	**Poor**	**Satisfactory**	**Good**	**Excellent**	**CLOs Mapping**
			0-25%	25%-50%	50-75%	75-100%	
R1	**Abstract**	5	Abstract is not written or written in a vague form	Abstract is written in an ordinary way	Abstract provides a good overview of project	Abstract is excellently written according to the scientific writing standards	CLO9
R2	**Problem Statement & Scope**	5	Problem is not stated at all or vaguely stated	Problem is stated but lacks necessary justification	Problem statement covers necessary Justification	Problem statement covers sufficient justification and reader can clearly understand its value and context	CLO6

R3	**Background, Literature Review & Analysis**	5	No or little background study of problem, market, and existing solutions	Organized but incomplete & inconsistent background study of problem, market, and existing solutions	Competent background study of problem, market, and existing solutions conducted and reported	Comprehensive and systematically executed background study of problem, market, and existing solutions	CLO2
R4	**Engineering Design/ Methodology**	5	The approach taken to solve the problem is not discussed No engineering design at all	Methodology is discussed with insufficient details Conventional design with new approach	Methodology is discussed with only major details Novel design with minimal impact	Methodology is discussed with major & minor details and supporting figures Novel design which solves complex engineering problem	CLO5
R5	**System/ Architecture Requirements**	5	System architecture/design is too vague	System design is included in an ordinary way	System design is included in an acceptable way	System design is included in an excellent way with details of components, setups, diagrams etc	CLO4
R6	**Implementatio n and Testing**	5	No system implementation and testing or very vague	System implementation is included in ordinary way. Testing is not adequate enough to test the entire system	System implementation is added in standard way, various cases are tested but quality of test cases can be improved	System implementation is added in standard way, all necessary cases are tested in scientific manner	CLO3
R7	**Results & Evaluations**	5	Results and evaluations are not provided	Results and evaluations are briefly discussed without supporting material	Results and evaluations are discussed with little supporting material	Results and evaluations are comprehensively discussed with sufficient supporting material	CLO4 CLO11
R8	**Conclusions & Future Work**	5	Conclusion does not summarize Project contributions, impact and limitations. No recommendations for future work	Largely qualitative conclusions rather than quantitative with limited discussion. Recommendatio ns for future work are not well thought out.	Conclusions are stated in quantitative manner and covers most aspects of projects. Recommendations for future work are properly planned.	Conclusions summarize all aspects of projects in quantitative as well as qualitative manner. Recommendations for future work are provided with suggested design process.	CLO11
R9	**Language, Grammar& Formatting**	5	Lot of spelling and grammatical mistakes and writing is not understandable with improper formatting	Frequent spelling grammatical mistakes and writing & formatting needs significant editing	Occasional spellings and grammatical mistakes and writing & formatting is acceptable	Almost no spelling or grammatical mistakes with excellent writing and formatting	CLO9

R10	**References & citations**	5	No or very few references and no citation in the text	Reasonable references but no citation in the text	Reasonable references and citation in the text but not in IEEE format	Reasonable references in IEEE format and well cited in the text	CLO7 CLO3
R11	**Plagiarism/ Similarity Index**	5	Most parts of the FYPreport is copied	More than 50% of the FYP report is copied	Most parts of the FYPreport is written by students	Similarity index of FYP report is in acceptable range	CLO7

(b) FYP Final Presentation Evaluation Rubrics

	Components	Marks	Poor	Satisfactory	Good	Excellent	CLOs Mapping
			0-2	2-4	5-8	8-10	
D1	**Problem Statement Scope & Sustainability**	10	Clear list of statements of problem is missing	Statements of problems are listed but does not reflect scope of project, thus need to be refined	List of statements of problem are clear and adequate for scope of project	List of statements of problem are concise and precise in relation to scope of project	CLO6
D2	**Background, Literature Review and Analysis**	10	No or little review of background and literature with no analysis	Review is conducted but incomplete and inconsistent with inappropriate analysis	Sufficient review conducted but not systematic with adequate analysis	Comprehensive and systematically executed review with thorough analysis	CLO2
D3	**Engineering Design/Approach**	10	No engineering design at all	Conventional design with new approach	Novel design with minimal impact	Novel design which solves complex engineering problem	CLO5
D4	**Results and Evaluations**	10	Results and evaluations are not provided	Results and evaluations are briefly discussed without supporting material	Results and evaluations are discussed with little supporting material	Results and evaluations are comprehensively discussed with sufficient supporting material	CLO4 CLO11
D5	**Engineering Knowledge**	10	Unable to answer basic questions	Answered some of rudimentary questions	Seems novice and can answer questions without elaboration	Answers to questions are strengthen by rationalization and explanation	CLO1
D6	**Presentation and Organization Skills**	10	Presentation is unstructured with key points are missing. The contents are hard to understand and interpret.	Presentation is articulated clearly with most of the key points covered. Difficult to follow due to lack of graphs, figures etc., to explain salient points.	Presentation is articulated clearly, the flow is reasonable with all the key points are covered but limited use of charts, graphs, figures etc., to explain salient points.	Presentation is articulated clearly, organized in a structured way with all key points are covered. Enhances presentation and keeps interest by effective use of charts, graphs, figures etc., to explain salient points.	CLO9
D7	**Timeline and Project Completion**	10	The project could not be completed	Major features of the project are completed but the timeline was not followed	Most features of the project were completed and timeline as defined in the proposal was followed	The project is completed with all features implemented according to the timeline defined in project proposal	CLO10
D8	**Professional Ethical Values**		Student is unaware of ethical implications of the work	Student is aware of ethical implications of the work but unable to ensures adherence to ethical best practices	Student is aware of ethical implications of the work and tried his best to ensures adherence to	the student has a good grasp on all the ethical implications of the work and ensures adherence to ethical best practices	CLO7

					ethical best practices		
D9	**Teamwork**	10	The student did not contribute to the project at any stage. Conflicts between the group members were clearly visible	The student contributed a little to the project as unable to answer basic questions	The student contributed mostly according to mentioned work division and cooperation between group is reasonable	The student contributed fully &actively according to mentioned work division and any conflicts within the group members were amicably resolved	CLO8

FYP Report:

- You should modify/extend your FYP Mid report in Latex on Overleaf to prepare FYP Final report. But you should save original .tex file of your proposal and mid for future.
- You FYP Final Report should have the following contents (Details in attached Rubrics):
    - Abstract
    - Introduction, Background, Scope & Need, sustainability, problem statement
    - Literature review, state-of-the Art, analysis based on background/literature review to figure out previous approaches/methods/designs used to solve the problem with their performance parameters. You can use comparison table, charts, graphs etc to show your comparison.
    - Engineering Design: your engineering design approach to solve the problem.
    - System/architecture requirements: Complete system design and architecture and its components and module requirements.
    - Implementation and Testing: How system/design is implemented and tested.
    - Results & Evaluations: Results obtained, results can be simulation or experimental results. You should analyze/evaluate/investigate your results, don't just report.
    - Conclusions and Future Work: conclude your work and highlight future directions.
    - References and Citations: You need to add and cite references in text as well as in figure captions taken from these references. You need to write in your own language, you can't just copy/paste from references in addition ti citations.
    - Plagiarism/similarity Index: You need to Add plagiarism report at the end of your report once your FYP report is finalized (reviewed and approved by advisor & co-advisor).
- **Plagiarism report**: It can be generated by emailing ITU library. Please email at reference.librarian@itu.edu.pk to get a report.
- **Plagiarism index**: Index should be below 19% and not exceed 5% for a single reference.
    - You don't have to include complete plagiarism report, you need to include first page of plagiarism report and originality report pages which shows similarity index, and all references. No need to include highlighted FYP report portion in the plagiarism report. An example of these two pages are as follows:

Example Turnitin Report

by Jane Doe

This extracts from an example Turnitin report will help you to understand how to read and interpret your own Turnitin Reports.
Turnitin highlights passages and phrases that it recognises as having come from other sources. However, not all highlighting suggests plagiarism, so it is important to review these reports alongside your referencing guidelines.

FILE	EXAMPLE_ESSAY.DOCX (43.4K)		
TIME SUBMITTED	14-JAN-2015 12:33PM	WORD COUNT	5689
SUBMISSION ID	39601934	CHARACTER COUNT	32004

Example Turnitin Report

ORIGINALITY REPORT

15% SIMILARITY INDEX | 9% INTERNET SOURCES | 1 PU

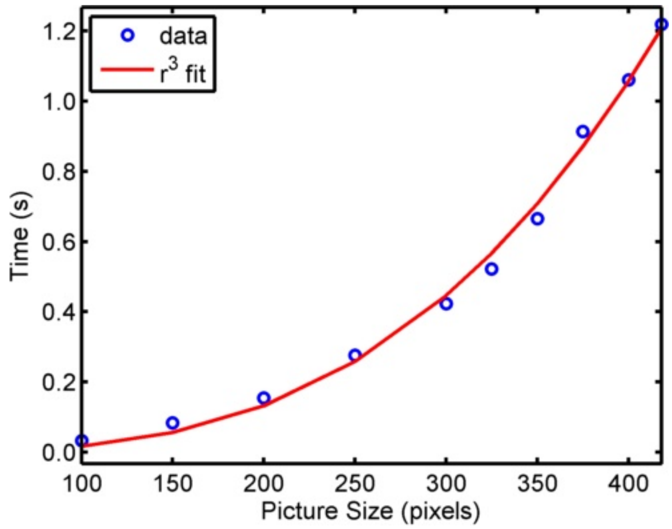
This number indicates what percentage of your essay can be found in other sources. This number *does not* indicate whether or not you have *plagiarised* work from other sources.
It is important that you look through your work carefully and make sure that you know how to reference the sources you use in the correct way for you discipline.

PRIMARY SOURCES

1	Mosley, J. E., and C. M. Grogan. "Representation in Nonelected Participatory Processes: How Residents Understand the Role of Nonprofit Community-based Organizations", Journal of Public Administration Research and Theory, 2012. Publication	2%
2	plato.stanford.edu Internet Source	1%
3	Rubenstein, Jennifer C.. "The Misuse of Power, Not Bad Representation: Why It Is Beside the Point that No One Elected Oxfam : Misuse of Power, Not Bad Representation", Journal of Political Philosophy, 2013. Publication	1%
4	Michael Saward. "Authorisation and Authenticity: Representation and the Unelected", Journal of Political Philosophy, 2/11/2008 Publication	1%
5	Submitted to London School of Economics and Political Science Student Paper	1%

This shows how much of your overall similarity rating is made-up of each individual source detected by Turnitin.

- Pay special attention to**Language, Grammar & Formatting**of report
- You should email you FYP final Report at least one week before deadline to you Advisor and Co-advisor for review. You should keep fyp.ee@itu.edu.pk in cc while emailing the report to your Advisor and Co-advisor.
- Advisor and co-advisor should sign the report with date after review and corrections if required. You have to upload the approved and signed FYP final report in pdf in response to this google classroom assignment (will be created).
- Name the proposal report as Group#00_Supervisor_Name_FYP_Report.pdf.
- See the above FYP report and presentation evaluation criteria for more details.
- You have to avoid following common mistakes in your FYP report:
    - grammar and sentence mistakes, you can install Grammar yapp in your chrome browser and Microsoft Word to check any grammatical errors.
    - The abbreviation of the text should be mentioned in full form when mentioning it for the first time followed by its acronym in the bracket. You can use the acronym afterwards.
    - The problem statement should be defined clearly even if it takes one page or more.
    - Quality and resolution of figures should be high, if you are copy/pasting some figures from literature , make sure it is of high quality, resolution and most important readable after pasting.
    - *You can't paste pictures of oscilloscopes and other data capturing tools, you have to export data from these sources and need to plot these in graph plotting tools like Matlab, Excel, Origin etc.*
    - Make sure that font and sizes of graphs axis labels and numbers should be readable and consistent throughout the report. One example of good graph is as follows:



    - *Section headings, figure captions, table captions etc should be meaningful.* We can understand that section headings have to be brief but there is no restriction that heading should be of one word, it can be of few words. Avoid section headings like Matlab, problem, working, result and many more like these, avoid figure caption like block diagram etc, block diagram of what?
    - It does not make any sense to make sections and paragraphs of 1, 2 or 3 lines.

- Tables can't be captioned as figures, you have to label them as tables. 10. Every table and figure should be referred and explained properly in the text i.e., in Table 1 or Figure 1.1 experimental results of XYZ are shown ......
- You should add your codes and programs in the **Appendix sections** after references section, avoid including these in the main report text.
- You should discuss and analyze your results in addition to just report these results.
- Everything taken from literature including figures and literature should be cited by proper reference numbers like [1] etc.
- *References should be in IEEE format*, examples are provided in the references section on how to include references for journal papers, books, book chapters, website etc. Most of you just provided the link of journal paper while citing the journal papers and format of citing journals, books, websites are totally wrong.
- There are many other similar mistakes, this is your report so you have to review and correct these mistakes/errors. Department will not accept your reports with errors and mistakes even if your reports are printed and signed. Sit with your advisors, co-advisors and other team members and improve the quality of your report in the light of above comments (there may be other kind of mistakes/errors).

FYP Defense Presentation:

- Every group member should be present at the presentation venue at least 10 minutes before scheduled time.
- Every member needs to present some portion of presentation and mode of presentation will be English.
- FYP defense presentation will be evaluated on the basis of Rubrics in the previous section, make sure to include these points in your presentations.
- You will have max. 10-15 minutes for your final presentation followed by Q&A session.
- You should include short and brief demo video/videos in your presentation.
- You will be uploading your FYP Defense Presentation after FYP Defense in response to assignment (will be created) on Google Classroom.
- You have to incorporate the changes in your presentation recommended by advisor and co-advisor before uploading it.
- Name the ppt/pptx file as Group#00_Advisor_Name_Defense.pptx

FYP Poster and Display Demonstration

FYP Display Demonstration Evaluation Rubrics

	Components	Marks	Poor	Satisfactory	Good	Excellent	CLOs Mapping

			0-2	2-4	5-8	8-10	
D1	**Completeness**	10	The system failed to produce the required results	The system produced inaccurate or incomplete results due to non-functional parts or incomplete implementation	The system produced most of the required results and most of the features were implemented	The system produced all the required results and all of the features were implemented. It was demonstrated how the real world problem was solved	CLO10
D2	**Scientific and Safety Standards**	10	Scientific and safety standard practices are not followed.	Scientific and safety standard practices are rarely followed	Scientific and safety standard practices are followed appropriately	Scientific and safety standard practices are followed extensively	CLO5
D3	**Demonstration & Display Style**	10	Students were unable to demonstrate the functional requirements at all	It is demonstrated how the system fulfills some of its functional requirements	It is demonstrated how the system fulfills most of its functional requirements but display can be improved	It is clearly and effectively demonstrated how the system fulfills all of its functional requirements with user friendly display	CLO9
D4	**Subject Knowledge**	10	Unable to answer basic questions	Answered some of rudimentary questions	Seems novice and can answer questions without elaboration	Answers to questions are strengthen by rationalization and explanation	CLO1
D5	**Originality & Creativity**	10	Working product is straightforward work with little to no creative potential	Working product has some potential for making a contribution to society	Working product has some creative/ original/ inventive element with potential for contribution to society	Working product has several creative/ original /inventive elements with clear potential for making contribution to society	CLO7 CLO11
D6	**Implementation and Modern Tool Usage**	10	Modern engineering tools were not used where applicable to solve complex engineering problems	Modern engineering tools were used rarely	Modern engineering tools were used but more could have been used to solve the problem	Modern engineering tools were used extensively and efficiently and new software/languages/ tools were learned as needed	CLO3

FYP Poster Evaluation Rubrics

	Components	Marks	Poor	Satisfactory	Good	Excellent	PLOs Mapping
			0-1	1-2	2-3	3-5	

P1	**Overall Appearance** RULE: Use a light background color, solid non-gradient fill pattern; 2 or 3 font colors; dark text on light background Don't create large, monolithic blocks of text, there should be text / graphics balance	5	Cluttered or sloppy appearance with too much or not enough text	Appearance is acceptable with no text graphics balance	Appearance is pleasant but not effective with enough text and graphics	Appearance is very pleasant and effective with text / graphics balance	CLO9
P2	**Organization & Flow** RULE: Use headings in contrasting color; use 3 or 4 column format for to indicate logical flow (top to bottom then L to R)	5	Unorganized with no logical flow	There is some organization but cannot figure out how to move through poster	Implicit flow used by making headings stand out (Methods, etc)	Explicit numbering used or columns used to indicate logical flow (top to bottom then L to R)	CLO9
P3	**Project Objectives** RULE: Tell readers why your work matters!!	5	Unable to find objective	Some unclear objectives are mentioned	Objectives are present but not convincing and explicit	Objectives are explicit and convincing	CLO9
P4	**Methodology** RULE: Tell reader about the design and development; tools technologies and methodologies used in project development	5	Unable to figure out methodology	Some unclear terms are shown and rest is left for users to figure out	Partial or incomplete information to comprehend method and main variables especially the outcome variables	Complete information which includes tools, technologies and methodologies With variables	CLO9
P5	**Results** RULE: Share the main and important results relevant to the research objectives/aims.	5	No results at all or irrelevant results	Few results are present	Results are present but not clear and explicit without used of figures and tables	Explicit results are present with proper headings, captions, figures etc	CLO9
P6	**Discussion/Conclusion /Recommendation** RULE: Interpret findings, summarize and recommend, what's next…	5	No conclusions and recommendations	Conclusions are just summary of results	Some conclusions and recommendations are provided but not well thought-out	Concise, analyzed and thought-out conclusions and recommendations are presented	CLO9

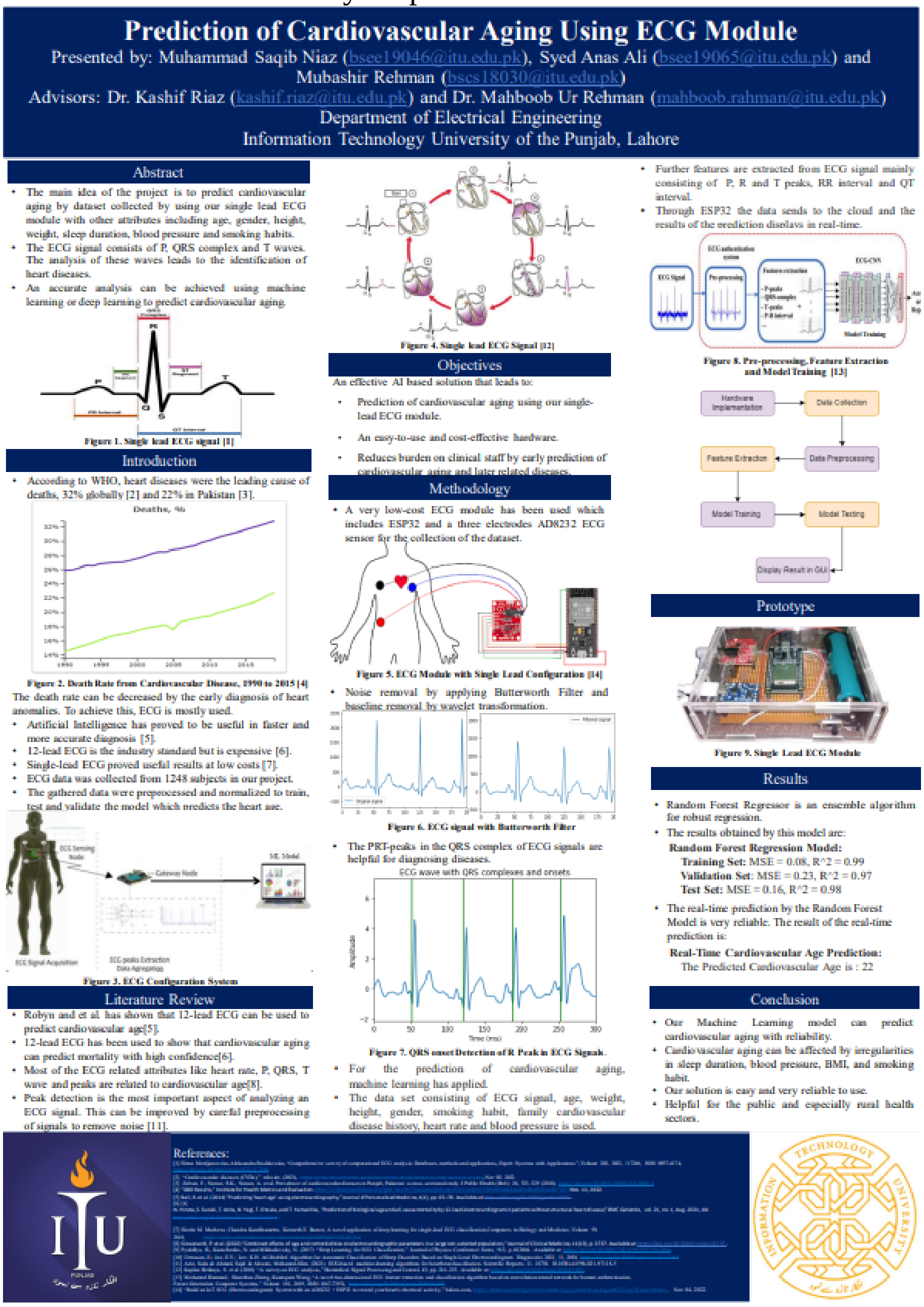
FYP Video:

- You have to prepare a video (max. 2 minutes) of your FYP clearly showing modules, setup, results and achievements of your FYP.
- You can edit your video to add voice or text in your FYP video.
- You have to get this video approved from your advisor by emailing him/her video, keep fyp.eed@itu.edu.pk in cc.
- Upload the approved video in response to assignment (will be created) on Google classroom.
- Name the video as Group#_Supervisor_Name_FYP_Video.

FYP Poster:

- You have to prepare a poster of your FYP clearly showing abstract, problem statement, methodology, results etc.
- You have to get this poster approved from your advisor by emailing him/her, keep fyp.eed@itu.edu.pk in cc.

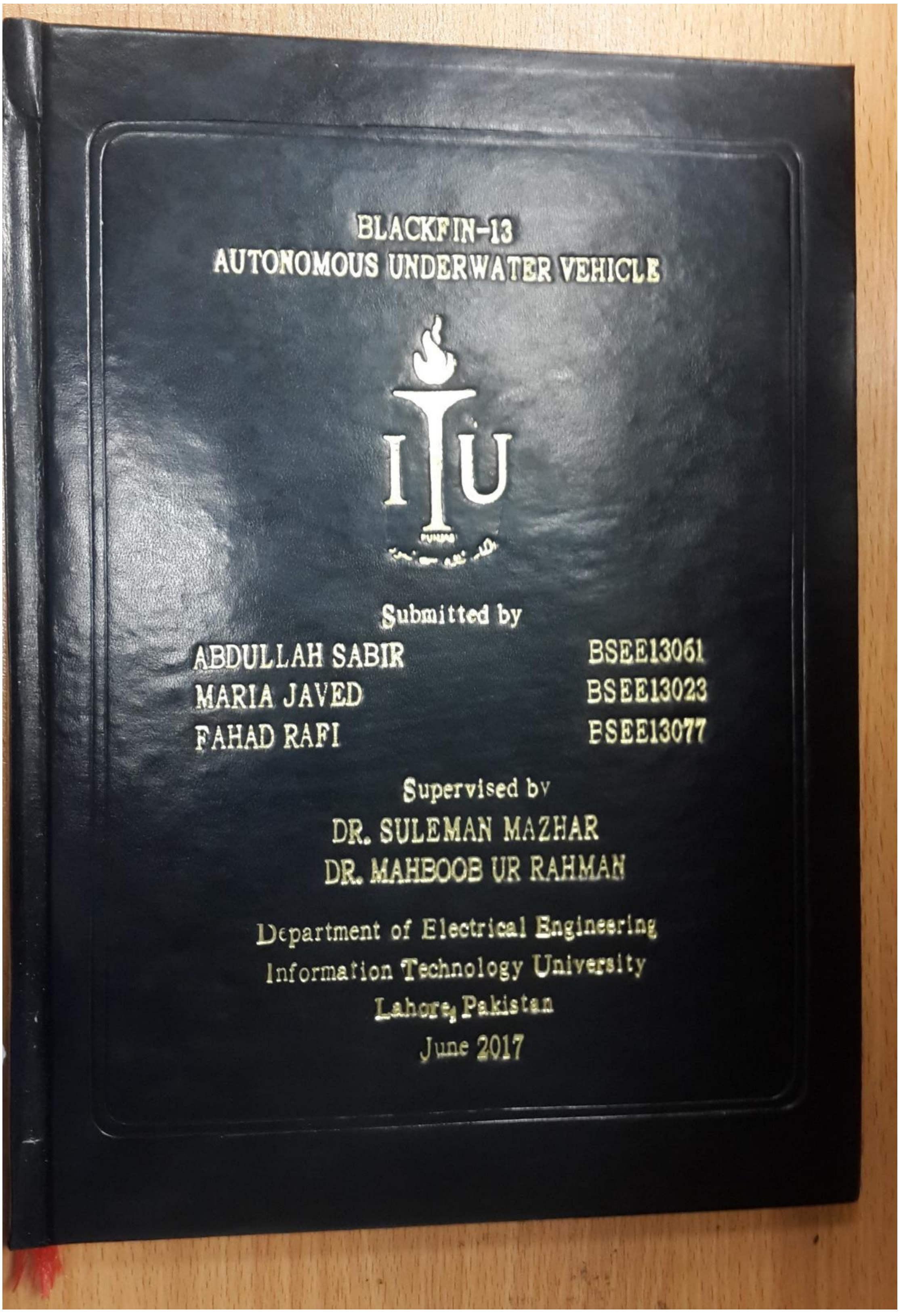
- Upload the approved poster in response to assignment (will be created) on Google classroom.
- Name the poster as Group#_Supervisor_Name_FYP_Poster.pptx
- You should avoid putting a lot of data in your posters, rather you should use bullet points in your poster.
- Obviously you are not bound to use the same format and style but you must include things mentioned in poster evaluation rubrics, your affiliation, advisors names, University logos etc.
- You can use A0 or A1 sizes for your poster.



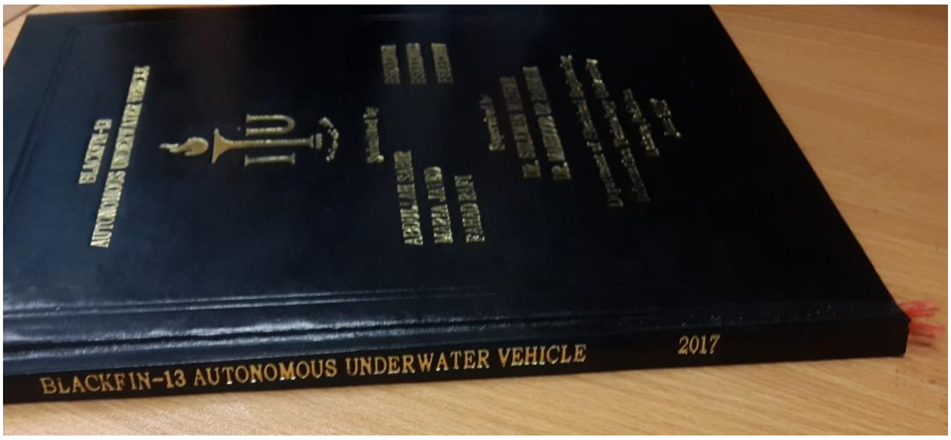
FYP Display Demonstration:

- FYP Live Display Demonstration will be scheduled and announced.
- FYP Groups have to setup their project in labs (or specified venue) for live demonstration (preferably Microwave and Control Labs on 4th Floor).
- FYP Display Evaluation Rubrics are mentioned above

Embossed Front View of Thesis



Embossed Side View of Thesis



Embossed Pages Template

Project Title

Submitted by

StudentName1 (BSEE130XX)

StudentName2 (BSEE130XX)

StudentName3 (BSEE130XX)

Supervised by

Advisor Name

Co-advisor Name

Department of Electrical Engineering

Information Technology University

Lahore, Pakistan

Month Year

Project Title

2024

REPORT TEMPLATE

- Consult provided Latex file for FYP Report template